

JONATAN EBENHOLM

PROFILE

Software / Systems Engineer

graduating June 2026, with strong C++, C# and Python fundamentals. Experience spans low-level systems and GPU programming, computer vision, simulation, and applied machine learning. Seeking software or systems engineering roles from summer 2026 — especially keen on graphics, GPU, computer vision, or simulation work.

CONTACT DETAILS

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PERSONAL INFORMATION

Citizenship: **Sweden**

Languages: **Swedish** (Native),
English (Fluent)

SKILLS

- C++20, C#, Python, MATLAB
- OpenGL 4.3+, GLSL/HLSL
- GPGPU, compute shaders
- TensorFlow, OpenCV
- Unity, Godot, Blender
- CMake, Git
- Agile / Scrum

EDUCATION

MSc Media Technology and Engineering

Linköping University, 2020–2026
Specialising in Computer Graphics, GPU programming, systems programming, and Machine Learning / AI.

PROJECTS

MINECRAFT TERRAIN ENGINE (C++20/OPENGL)

2025–2026

- GPU-driven voxel engine using OpenGL 4.3+ compute shaders with indirect drawing for high-throughput rendering.
- Multi-threaded chunking architecture with spline-based terrain shaping for an infinite, seamless world.

GPU PATH TRACER (C++/OPENGL)

2025

- Physically-based path tracer (reflections, refractions, Lambertian surfaces, area lights) implemented in a fragment shader.
- CPU-built BVH traversed via Shader Storage Buffer Objects; handles tens of thousands of triangles.

GESTURE CONTROLLER (PYTHON / TENSORFLOW)

2024

- Real-time gesture recognition driving a Godot game over Web-Socket: OpenCV capture, MediaPipe hand tracking, a custom CNN classifier.
- Published as a research paper comparing CNN vs. feed-forward approaches on landmark data.

ELEMENTAL CLASH (UNITY / C#)

2023

- Shipped a 1v1 real-time strategy game for a real client as a team of seven, using agile methods (Scrum) from spec to handoff.
- Acted as **Product Owner** — owned the client relationship, requirements, and delivery expectations.
- Integrated ArUco marker-based computer vision (OpenCV) for physical card placement as the core input mechanic.

EXPERIENCE

MASTER THESIS — INVERSE RENDERING FOR INDUSTRY INSPECTIONS

SICK IVP / Linköping University

2026.01–2026.06

- Combining computer graphics, deep learning, and image processing to recover surface maps for automated defect detection.
- Building the full data collection, model evaluation, and real-world validation pipeline on-site.

PROGRAMMING ASSISTANT — LITHEHACK at Linköping University

2025.09–2026.06

- Supported students with programming problems, debugging, and core CS concepts.
- Nominated for the role by peers — not applied for.

LAB ASSISTANT — APPLIED TRANSFORM THEORY at Linköping University

2024.01–2024.03

- Guided students through lab sessions on Fourier, Laplace, and Z-transforms using MATLAB.
- Scouted for the role by the course examiner.